

Operating Systems - Unit 1: Process Management

A process is an instance of a computer program that is being executed by one or many threads.

Key states: New, Ready, Running, Waiting, Terminated.

Chapter 3: Deadlock Prevention and Avoidance

Four Coffman conditions: Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait.

Banker's algorithm ensures dynamic avoidance by verifying safe states.

Chapter 4: Memory Virtualization

Paging eliminates external fragmentation through fixed-size physical frames.